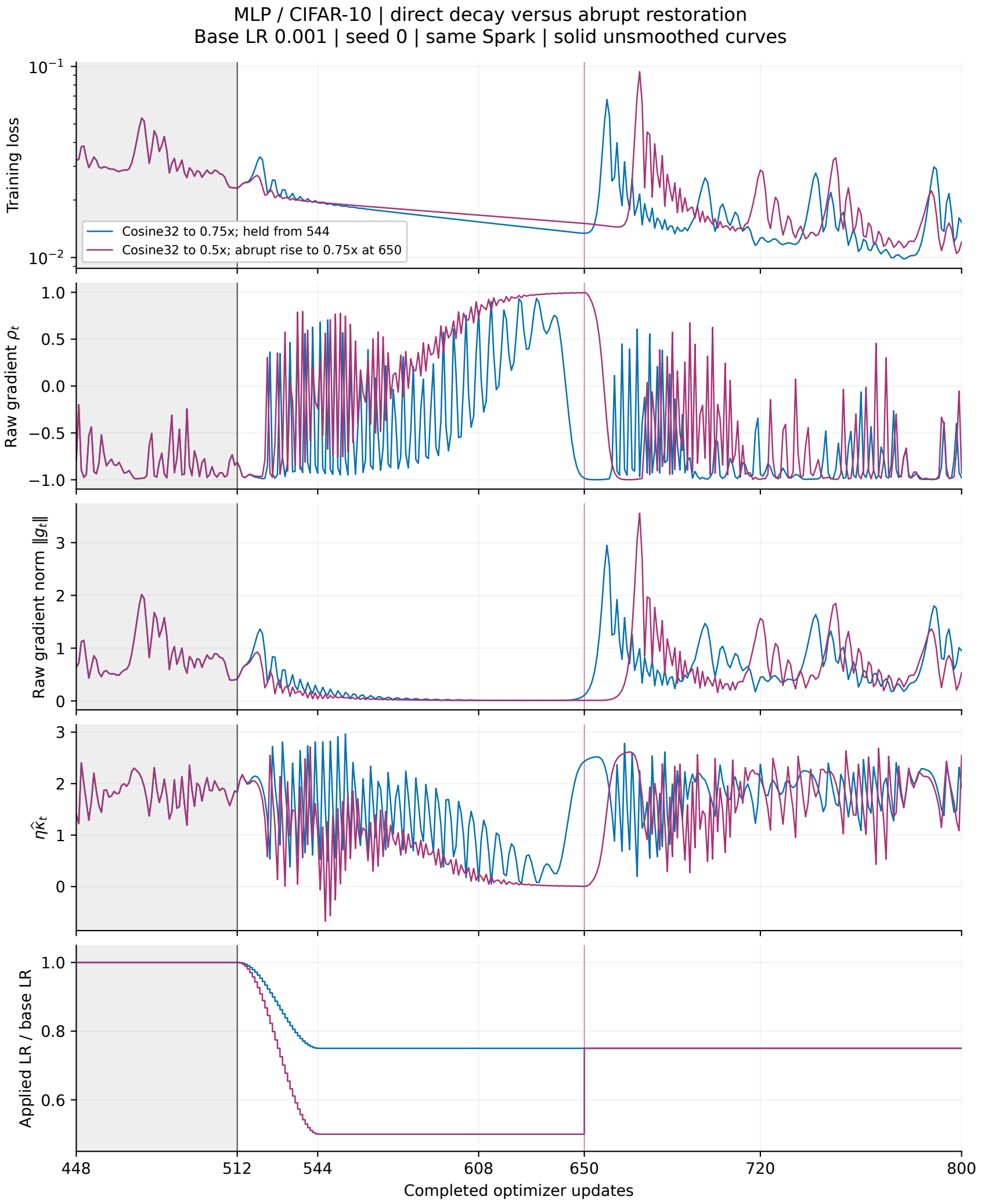

MLP / CIFAR-10 | direct decay versus abrupt restoration
Base LR 0.001 | seed 0 | same Spark | solid unsmoothed curves
Training loss
Cosine32 to 0.75x; held from 544
Cosine32 to 0.5x; abrupt rise to 0.75x at 650
Raw gradient ρ_t
Raw gradient norm $\|g_t\|$
$\eta\hat{\kappa}_t$
Applied LR / base LR
Completed optimizer updates